

# Introduction

*Benoît Legat*

*Understand simple things deeply*

☐ Full Width Mode    ☐ Present Mode

## ☰ **Table of Contents**

**Deepseek event**

**What do these libraries really do ?**

**Grading**

# Deepseek event ⇄

## Nvidia stocks recovered slightly on Tuesday

Nvidia share price



Note: Axis does not start at zero. Markets are open Mon-Fri 9:30-16:00 Eastern time

Source: Bloomberg. Last update: 28 Jan 2025

B B C

# Distillation ⇄

## 2.3.1. Cold Start

Unlike DeepSeek-R1-Zero, to prevent the early unstable cold start phase of RL training from the base model, for DeepSeek-R1 we construct and collect a small amount of long CoT data to fine-tune the model as the initial RL actor. To collect such data, we have explored several approaches: using few-shot prompting with a long CoT as an example, directly prompting models to generate detailed answers with reflection and verification, gathering DeepSeek-R1-Zero outputs in a readable format, and refining the results through post-processing by human annotators.

In this work, we collect thousands of cold-start data to fine-tune the DeepSeek-V3-Base as the starting point for RL. Compared to DeepSeek-R1-Zero, the advantages of cold start data

| Model                        | AIME 2024 |         | MATH-500 | GPQA Diamond | LiveCodeBench |
|------------------------------|-----------|---------|----------|--------------|---------------|
|                              | pass@1    | cons@64 | pass@1   | pass@1       | pass@1        |
| QwQ-32B-Preview              | 50.0      | 60.0    | 90.6     | 54.5         | 41.9          |
| DeepSeek-R1-Zero-Qwen-32B    | 47.0      | 60.0    | 91.6     | 55.0         | 40.2          |
| DeepSeek-R1-Distill-Qwen-32B | 72.6      | 83.3    | 94.3     | 62.1         | 57.2          |

Table 6 | Comparison of distilled and RL Models on Reasoning-Related Benchmarks.

## Low-level improvements ⇄

.....

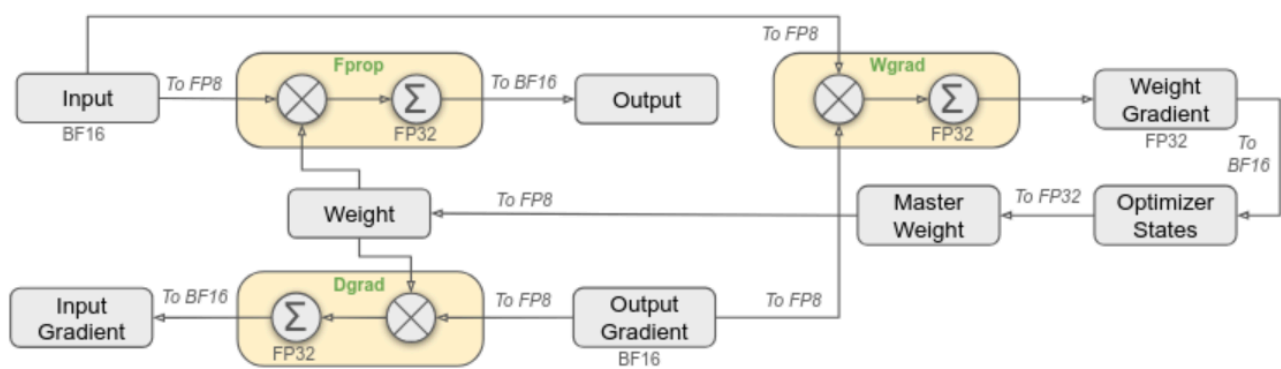


Figure 6 | The overall mixed precision framework with FP8 data format. For clarification, only the Linear operator is illustrated.

► Is it worth developing such low-level improvements?

# What do these libraries really do ? ⇄



## Error message from ArgTools

```
RequestError: HTTP/2 404 while requesting https://www.gstatic.com/devrel-  
devsite/prod/vf0eb6f8ebad49d1e3523c3ee6bef7563a09624802753ccc3d816c9b276e850fd/tensor  
flow/images/lockup.svg
```

## Stack trace

Here is what happened, the most recent locations are first:

1. anonymous function  
from `Downloads.jl:285`
2. `#open#324(f::Downloads.var"#7#8"{...}, args::String; kwargs::@Kwargs{...})` ...show types...  
from `io.jl:429`
3. `#open_nolock#1(::Function, ::Vararg{...}; kws::@Kwargs{...})` ...show types...  
from `ArgTools → ArgTools.jl:35`
4. `open_nolock`  
from `ArgTools.jl:35`
5. `arg_write(f::Function, arg::String)`  
from `ArgTools → ArgTools.jl:103`

6. #download#5

from Downloads.jl:272

7. download

from Downloads.jl:252

8. #RobustLocalResource#30(url::String, path::String, html\_attributes::Pair{...}; cache::Bool) ...show types...

from robustlocalresource.jl:20

9. RobustLocalResource

from robustlocalresource.jl:7

10. #save\_image#2

from Other cell: line 24

```
22 function save_image(url::URL, html_attributes...; name = split(url.url,
'/'')[end], kws...)
23     path = joinpath("cache", name)
24     return PlutoTeachingTools.RobustLocalResource(url.url, path, html_attr
ibutes...), path
25 end
26
```

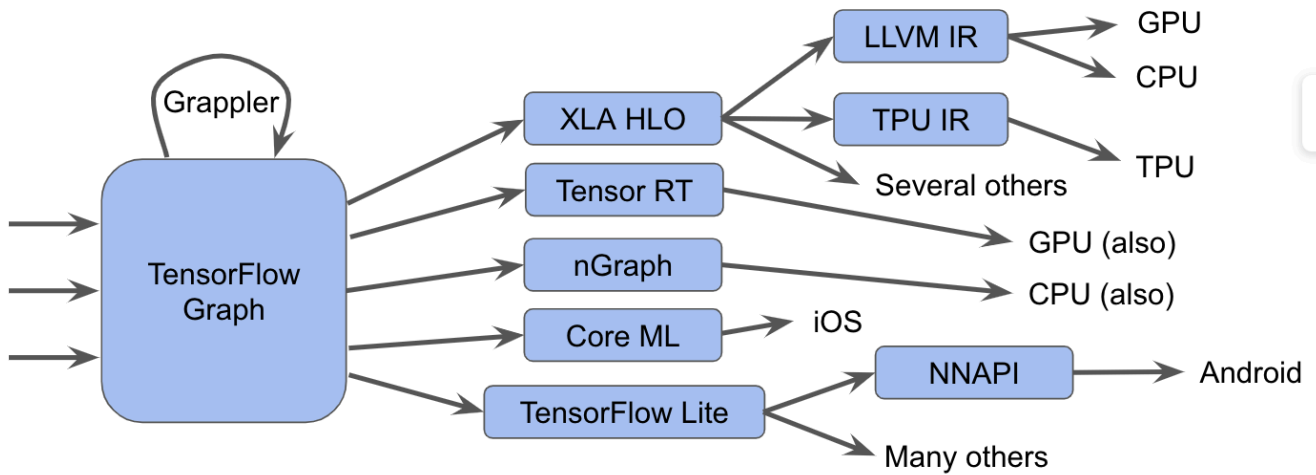
cell preview

11. Show more...

```
1 img("https://www.gstatic.com/devrel-
devsite/prod/vf0eb6f8ebad49d1e3523c3ee6bef7563a09624802753ccc3d816c9b276e850fd/tenso
rflow/images/lockup.svg", :height => 100)
```

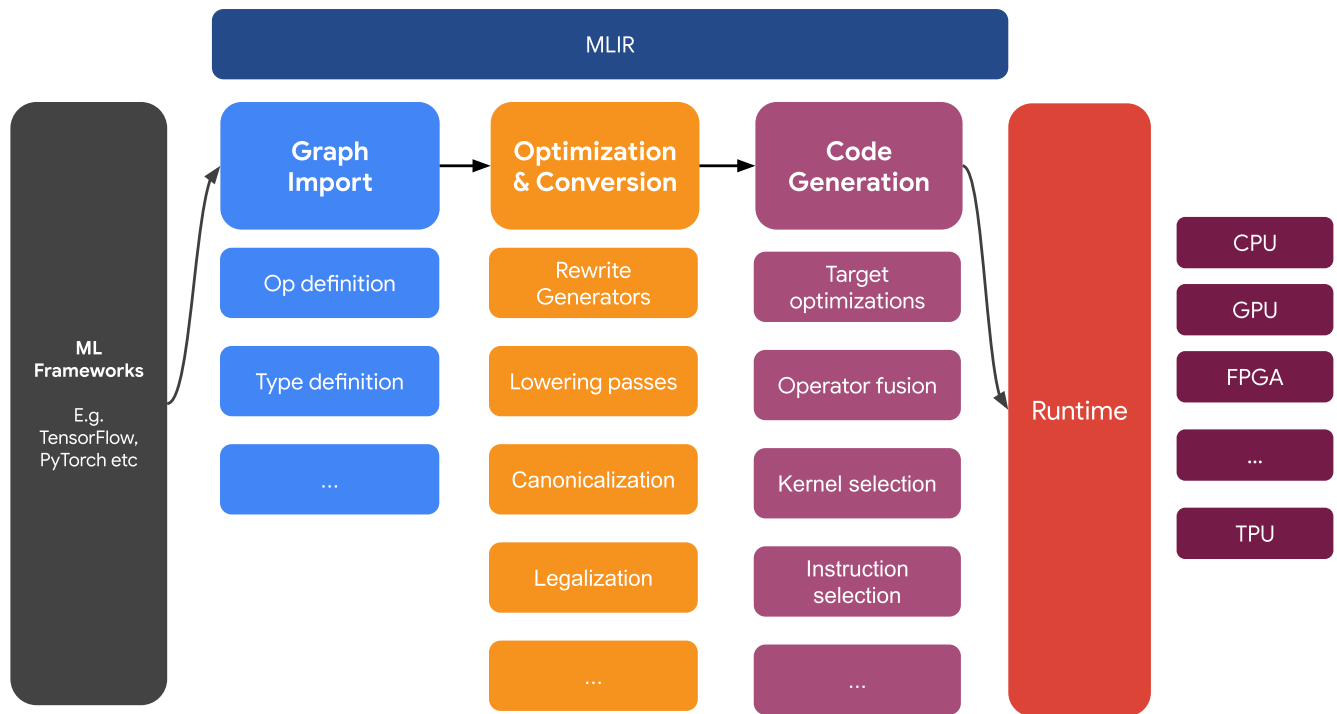
👁 Reading hidden code

# Accelerated automatic differentiation ⇄



## MLIR ↗

---



## DIY ↗

---

The goal of the course is to take a tour of **automatic differentiation**. For this we will write our own from scratch, in **julia**.

Yann LeCun 

@ylecun · [Follow](#)



Hotter take: ML would have advanced faster if another front-end language had been available and widely adopted instead of Python.

One that is interactive yet fast & compilable, multithreaded (no GIL), isn't bloated, doesn't care about white spaces,... E.g. Julia or some Lisp.

Bojan Tunguz  @tunguz

Hot take: Machine Learning would not have been nearly as advanced now were it not for Python. Python's two main virtues in the context of ML:

1. Lowering barriers to entry.
2. As a scripting language, it encourages and enables experimental workflow.

1:27 PM · Feb 22, 2023



3.1K



Reply



Copy link

[Read 297 replies](#)



# Grading ⇄

---

- If exam or homework is below 5, grade is the minimum of both
- If exam and homework are above 10, grade is the average of both
- Otherwise, we interpolate between these cases as follows
- Gain bonus points by contributing to the Git or winning benchmarks with the projects

f (generic function with 1 method)

```
1 f(x) = min(x/20, 0.5)
```

g (generic function with 1 method)

```
1 g(a, b) = f(a) * b + (1 - f(a)) * a
```

grade (generic function with 1 method)

```
1 grade(HW, EX) = min(g(HW, EX), g(EX, HW))
```



The End

```
1 import PlotlyLight
```

```
1 using Colors, PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral,  
    PlutoTeachingTools, ShortCodes
```

img (generic function with 3 methods)

qa (generic function with 2 methods)